Computer Engineering Case Study

- Arithmetic Operations Learning Tool

Zichen Jing (Nancy) zj2256

ENGR-UH 1000, Spring 2023 (Assignment 2)

Step 1: Problem Identification and Statement

This assignment aims to develop a program with the use of functions and arrays, to help an elementary school student learn multiplication or division of positive one-digit integer numbers by asking them questions.

Step 2: Gathering Information and input/Output diagram

- *Gathering of information*

Arithmetic operations involved in this program are multiplication and division. All the questions and answers are in integer form.
For multiplication, the answer can only be integers when the factors are integers. Therefore, no special caution when answering multiplication questions. For example, 4 multiplied by 5 will give 20 as the answer.
For division, all the division equations will be asking for an answer in the quotient form for the answer to be an integer, excluding the reminder. The equation will be given in a form of x/y (x divided by y). The user will need to answer how many y is there in x. This means, for example, 8 divided by 5 will give 1 as the answer, meaning that there is one 5 in 8. Whereas 1/5 will give the answer of zero since there is no 5 in 1.
After the performance of the program, the user will be notified how many questions out of 10 questions he/she has answered correctly. If above 7, it means the user can skillfully carry out multiplication or division calculations. If below 7, the user might need to seek some help.

- *Input/Output Description*

Input:
The inputs are the selection of the number in the menu displayed on the screen

1. Input '1' if the user wants to perform multiplication

- Further input the answers of the randomly generated multiplication equations by the program in the form of integers. 10 questions will be displayed and therefore input 10 answers.

2. Input ‘2’ if the user wants to perform division
   - Further input the answers of the randomly generated division equations by the program in the form of integer quotient only, without reminder. 10 questions will be displayed and therefore input 10 answers.
3. Input ‘3’ to exit, which terminates the program
4. Any input other than 1, 2, or 3 will be seen as an error

Output:

1. A menu is displayed as output with three options
2. Corresponding to input 1

Display 10 multiplication equations each with two randomly generated positive one-digit integer numbers. Display a summary including the multiplication equations, and the user’s answers, mark the user’s answers with correct or incorrect (if incorrect, display the correct answer), correction rate in the form of x/10, and different messages for if the user answered more than 7 questions or less (including 7) than 7 questions correctly.

3. Corresponding to input 2

Display 10 division equations each with two randomly generated positive one-digit integer numbers. Display a summary including the division equations, and the user’s answers, mark the user’s answers with correct or incorrect (if incorrect, display the correct answer), correction rate in the form of x/10, and different messages for if the user answered more than 7 questions or less (including 7) than 7 questions correctly.

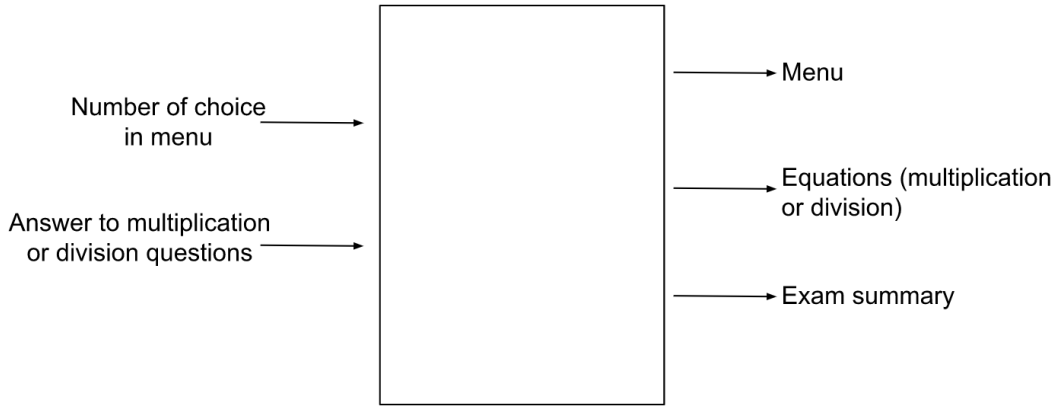
4. Corresponding to input 3

Display a goodbye message and then terminates the program.

5. Corresponding to any other input

An error message will be displayed. Prompt the user for another value.

Inputs and outputs are also shown in the diagram below:



Step 3: Test cases and Algorithm Design

- *Test Cases*

Sample of the main menu

Please choose one of the following: 1) Multiplication 2) Division (quotient) 3) Exit

Test Case 1: Invalid input in the menu selection

In the menu, users need to select the function they wish the program to do. If the user enters an invalid input (numbers other than 1, 2, or 3), the program will display an error message and prompt the user for another valid value.

Test Case 2: Selecting 1) Multiplication
with more than 7 correct answers

If the user selected 1) Multiplication, the program should display 10 multiplication equations one by one and prompt the user for the answers. After all the equations have been displayed and answered, the program will display an exam summary. The summary will include all the equations along with the user's answers, whether the answer is correct or not, and if the answer is incorrect, the correct answer will be displayed. Finally, a correction rate will be displayed in the form of x/10. In this case, more than 7 of the answers are correct, a "Congratulations" message will be displayed. The program will then direct the user back to the main menu to re-select the user's desired function.

Test Case 3: Selecting 1) Multiplication
with less than 7 correct answers (including 7)

Continuing from test case 2.
In this case, less than 7 (including 7) of the answers are correct, a "Please ask your teacher for help!" message will be displayed. The program will then direct the user back to the main menu to re-select the user's desired function.

Test Case 4: Selecting 2) Division (quotient)
with more than 7 correct answers

If the user selected 2) Division (quotient), the program should display 10 division equations one by one and prompt the user for the answers. After all the equations have been displayed and answered, the program will display an exam summary. The summary will include all the

equations along with the user's answers, whether the answer is correct or not, and if the answer is incorrect, the correct answer will be displayed. Finally, a correction rate will be displayed in the form of x/10. In this case, more than 7 of the answers are correct, a "Congratulations" message will be displayed. The program will then direct the user back to the main menu to re-select the user's desired function.

Test Case 5: Selecting 2) Division (quotient)
with less than 7 correct answers (including 7)

Continuing from test case 4.
In this case, less than 7 (including 7) of the answers are correct, a "Please ask your teacher for help!" message will be displayed. The program will then direct the user back to the main menu to re-select the user's desired function.

Test Case 6: Selecting 3) Exit

If the user selected 3) Exit, the program must display a message, such as "Thank you for using the program, goodbye!". And then quit the program.

- *Algorithm Design*

```
int print_Menu( )
    Declare integer choice;
    Print "Please choose the calculation you want to practice", newline
    Print "1) Multiplication", newline
    Print "2) Division (quotient)", newline
    Print "3) Exit", newline
    Print "Please make a selection ", newline
    Read value into choice
    While choice<1 or choice>3
            Print "The number you entered is invalid, please try again", newline
            Print "Please make a selection ", newline
            Read value into choice
    Return choice

int rand_int( )
    Return a random number between 1 and 9

void run_Test(pass integers: x[ ], y[ ], answers[ ], operation, size)
        Print newline
        Print "Please give the answers to the following multiplication questions in the form of
        integers", newline
```

```
    If operation=1
      Declare integer i and initialize to 0, while i<size, repeat the following, increment i
                        Set the seed for the random number generator to time
                        Assign rand_int( ) to x[i]
                        Assign rand_int( ) to y[i]
                        Print x[i], "*", y[i], "="
                        Read value into answers[i]
    If operation=2
      Declare integer i and initialize to 0, while i<size, repeat the following, increment i
                        Set the seed for the random number generator to time
                        Assign rand_int( ) to x[i]
                        Assign rand_int( ) to y[i]
                        Print x[i], "/", y[i], "="
                        Read value into answers[i]

int print_Summary(pass integers: x[ ], y[ ], answers[ ], operation, size)
  Declare integer ctr and initialize to 0
  If operation=1
    Print "Exam Summary:", newline
    Declare integer i and initialize to 0, while i<size, repeat the following, increment i
            Print x[i], "*", y[i], "=", answers[i]
            If x[i]*y[i]=answers[i]
              Print " ", "- correct answer", newline
              Increment ctr
            Otherwise
                    Print " ", "- incorrect answer – the answer is", " ", x[i]*y[i], newline
  If operation=2
    Print "Exam Summary:", newline
    Declare integer i and initialize to 0, while i<size, repeat the following, increment i
            Print x[i], "/", y[i], "=", answers[i]
            If x[i]/y[i]=answers[i]
              Print " ", "- correct answer", newline
              Increment ctr
            Otherwise
                    Print " ", "- incorrect answer – the answer is", " ", x[i]/y[i], newline
    If ctr>7
      Print "Congratulations! You scored ", ctr, "/10", newline
    Otherwise
            Print "You scored ", ctr, "/10. Please ask your teacher for help!", newline
```

Return ctr

Main function:
Define size as 10

Declare integers: operation(initialize to 0), array x[initialize to size], array y[initialize to size], array answers[initialize to size], choice(initialize to 0)

Repeat the following while choice≠3
Assign print_Menu( ) to choice

If choice=1
Assign choice to operation
Call the function run_Test(pass integers: x, y, answers, operation, size)
Print newline
Call the function print_Summary(pass integers: x, y, answers, operation, size)
Print newline
Break
If choice=2
Assign choice to operation
Call the function run_Test(pass integers: x, y, answers, operation, size)
Print newline
Call the function print_Summary(pass integers: x, y, answers, operation, size)
Print newline
Break

If choice=3
Print "Thank you for using the program, goodbye!", newline
Exit with 1

Step 4: Code of implementation

```
/*---------------------------------*/
/*Name: Zichen Jing (Nancy), NetID: zj2256 */
/*        Date: April 4, 2023        */
/*         Assignment 2.cpp         */
/*---------------------------------*/

/*----------------------------------------START-------------------------------------
-------*/

#include <iostream>
#include <iomanip>
```

```
#include <cstdlib>
#include <cmath>

using namespace std;

//function prototypes:

int print_Menu();

int rand_int();

void run_Test(int x[], int y[], int answers[], int operation, int size);

int print_Summary(int x[], int y[], int answers[], int operation, int size);


//function to display the menu for the user to choose
int print_Menu(){
    int choice;
    cout<<"Please choose the calculation you want to practice"<<endl;
    cout<<"1) Multiplication"<<endl;
    cout<<"2) Division (quotient)"<<endl;
    cout<<"3) Exit"<<endl;
    cout<<"Please make a selection: "<<endl;
    cin>>choice;
    //check the validity of the user's input, must be one of the numbers provided
    while (choice<1||choice>3){
        cout<<"The number you entered is invalid, please try again"<<endl;
        cout<<"Please make a selection: "<<endl;
        cin>>choice;}

    return choice;
}

//function to generate a random number which will be used in building the equations
int rand_int(){
    return rand()%9+1;
}

//function to display 10 multiplication or 10 division equations based on the user's
choice
void run_Test(int x[], int y[], int answers[], int operation, int size){
    cout<<"\n"<<endl; //empty line to make the statements displayed less crowded
    cout<<"Please give the answers to the following multiplication questions in the
    form of integers"<<endl;
    //if the user chooses to do multiplications
    if (operation==1){
        for (int i=0; i<size; i++){
            //set the seed of the random number generator
            srand((unsigned)time(NULL));
            //store into x array and y array
            x[i]=rand_int(); //call the random integer generating function
            y[i]=rand_int(); //call the random integer generating function
            //output the generated numbers into an equation form
            cout<<x[i]<<"*"<<y[i]<<"=";
            //store into answer array
            cin>>answers[i];
        }
    }
    //if the user chooses to do divisions
    if (operation==2){
```

```
        for (int i=0; i<size; i++){
            //set the seed of the random number generator
            srand((unsigned)time(NULL));
            //store into x array and y array
            x[i]=rand_int(); //call the random integer generating function
            y[i]=rand_int(); //call the random integer generating function
            //output the generated numbers into an equation form
            cout<<x[i]<<"/"<<y[i]<<"=";
            //store into answer array
            cin>>answers[i];
        }
    }
}

//function to display the summary of the user's answers, including the correct rate
and if needed, correction to wrong answers
int print_Summary(int x[], int y[], int answers[], int operation, int size){
    int ctr(0); //the number of correct answers
    //if the user chooses to do multiplications
    if (operation==1){
        cout<<"Exam Summary:"<<endl;
        //print the equations and the user's answers
        for(int i=0; i<size; i++){
            cout<<x[i]<<"*"<<y[i]<<"="<<answers[i];
            //if the user answered correctly
            if(x[i]*y[i]==answers[i]){
                cout<<" "<<"- correct answer"<<endl;
                ctr++; //increase the number of correct answers
            }
            //correction to wrong answers
            else
                cout<<" "<<"- incorrect answer - the answer is"<<" "<<x[i]*y[i]<<endl;
        }
    }
    //if the user chooses to do divisions
    if (operation==2){
        cout<<"Exam Summary:"<<endl;
        //print the equations and the user's answers
        for(int i=0; i<size; i++){
            cout<<x[i]<<"/"<<y[i]<<"="<<answers[i];
            //if the user answered correctly
            if(x[i]/y[i]==answers[i]){
                cout<<" "<<"- correct answer"<<endl;
                ctr++; //increase the number of correct answers
            }
            //correction to wrong answers
            else
                cout<<" "<<"- incorrect answer - the answer is"<<" "<<x[i]/y[i]<<endl;
        }
    }
    //answering correctly for more than 7 questions
    if (ctr>7)
        cout<<"Congratulations! You scored "<<ctr<<"/10"<<endl;
    //answering correctly for less than 7 questions
    else
        cout<<"You scored "<<ctr<<"/10. Please ask your teacher for help!"<<endl;

    return ctr;
}
```

```
int main() {

#define size 10 //declare "size" as 10

    //declare variables, initialized to values that will never naturally appear in the
      program
    int operation(0), x[size], y[size], answers[size], choice(0);

    do{
        choice=print_Menu(); //call the function to display the menu

        switch(choice){
            //when the user chooses to do multiplication
            case 1:
                operation=choice; //assign the user's choice of the menu to operation
                                    so that the function can respond correctly
                run_Test(x, y, answers, operation, size); //call the function to
                                                            generate equations based on
                                                            the user's choice
                cout<<"\n"<<endl; //empty line to make the statements displayed less
                                    crowded
                print_Summary(x, y, answers, operation, size); //call the function to
                                                                 display the summary of
                                                                 the user's progress
                cout<<"\n"<<endl; //empty line to make the statements displayed less
                                    crowded
                break;

            //when the user chooses to do division
            case 2:
                operation=choice; //assign the user's choice of the menu to operation
                                    so that the function can respond correctly
                run_Test(x, y, answers, operation, size); //call the function to
                                                            generate equations based on
                                                            the user's choice
                cout<<"\n"<<endl; //empty line to make the statements displayed less
                                    crowded
                print_Summary(x, y, answers, operation, size); //call the function to
                                                                 display the summary of
                                                                 the user's progress
                cout<<"\n"<<endl; //empty line to make the statements displayed less
                                    crowded
                break;

            //when the user chooses to exit the program
            case 3:
                cout<<"Thank you for using the program, goodbye!"<<endl;
                exit(1);
        }
    }while (choice!=3); //only way to terminate the program is by choosing option 3


    return 0;
}
```

Step 5: Test and Verification

Menu display

```
Please choose the calculation you want to practice
1) Multiplication
2) Division (quotient)
3) Exit
Please make a selection:
```

Test Case 1: Invalid input in the menu selection

```
Please choose the calculation you want to practice
1) Multiplication
2) Division (quotient)
3) Exit
Please make a selection:
4
The number you entered is invalid, please try again
Please make a selection:
```

(The user's input is 4, which is invalid, an error message is displayed, and asked the user for another valid value.)

Test Case 2: Selecting 1) Multiplication with more than 7 correct answers

```
Please choose the calculation you want to practice
1) Multiplication
2) Division (quotient)
3) Exit
Please make a selection:
1


Please give the answers to the following multiplication questions in the form of integers
1*3=3
5*1=5
9*8=72
3*2=6
7*9=63
6*4=24
1*2=2
5*9=45
4*5=20
8*3=24


Exam Summary:
1*3=3 - correct answer
5*1=5 - correct answer
9*8=72 - correct answer
3*2=6 - correct answer
7*9=63 - correct answer
6*4=24 - correct answer
1*2=2 - correct answer
5*9=45 - correct answer
4*5=20 - correct answer
8*3=24 - correct answer
Congratulations! You scored 10/10


Please choose the calculation you want to practice
1) Multiplication
2) Division (quotient)
3) Exit
Please make a selection:
```

(The user's input is 1 to perform multiplications. 10 multiplication questions are displayed in the form of x*y= for the user to input answers. The exam summary is displayed marking the user's answers, and displaying the correct answer if the user has answered incorrectly. Message in the case of more than 7 correct answers is displayed. Automatically directed back to the menu.)

Test Case 3: Selecting 1) Multiplication with less than 7 correct answers (including 7)

```
Please choose the calculation you want to practice
1) Multiplication
2) Division (quotient)
3) Exit
Please make a selection:
1


Please give the answers to the following multiplication questions in the form of integers
4*5=20
8*3=23
7*8=55
6*4=24
1*2=2
5*8=40
9*6=53
8*2=16
3*9=27
2*5=10


Exam Summary:
4*5=20 - correct answer
8*3=23 - incorrect answer - the answer is 24
7*8=55 - incorrect answer - the answer is 56
6*4=24 - correct answer
1*2=2 - correct answer
5*8=40 - correct answer
9*6=53 - incorrect answer - the answer is 54
8*2=16 - correct answer
3*9=27 - correct answer
2*5=10 - correct answer
You scored 7/10. Please ask your teacher for help!


Please choose the calculation you want to practice
1) Multiplication
2) Division (quotient)
3) Exit
Please make a selection:
```

(The user's input is 1 to perform multiplications. 10 multiplication questions are displayed in the form of x*y= for the user to input answers. The exam summary is displayed marking the user's answers, and displaying the correct answer if the user has answered incorrectly. Message in the case of less or equal to 7 correct answers is displayed. Automatically directed back to the menu.)

Test Case 4: Selecting 2) Division (quotient) with more than 7 correct answers

```
Please choose the calculation you want to practice
1) Multiplication
2) Division (quotient)
3) Exit
Please make a selection:
2


Please give the answers to the following division questions in the form of integers (quotient only, no reminder)
1/2=0
9/7=1
4/5=0
8/3=2
3/1=3
2/5=0
6/3=2
1/1=1
5/8=1
3/8=0


Exam Summary:
1/2=0 - correct answer
9/7=1 - correct answer
4/5=0 - correct answer
8/3=2 - correct answer
3/1=3 - correct answer
2/5=0 - correct answer
6/3=2 - correct answer
1/1=1 - correct answer
5/8=1 - incorrect answer - the answer is 0
3/8=0 - correct answer
Congratulations! You scored 9/10


Please choose the calculation you want to practice
1) Multiplication
2) Division (quotient)
3) Exit
Please make a selection:
```

(The user's input is 2 to perform divisions. 10 division questions are displayed in the form of x/y= *for* the user to input answers. The exam summary is displayed marking the user's answers, and displaying the correct answer if the user has answered incorrectly. Message in the case of more than 7 correct answers is displayed. Automatically directed back to the menu.)

Test Case 5: Selecting 2) Division (quotient) with less than 7 correct answers (including 7)

```
Please choose the calculation you want to practice
1) Multiplication
2) Division (quotient)
3) Exit
Please make a selection:
2


Please give the answers to the following division questions in the form of integers (quotient only, no reminder)
8/1=8
3/8=0
7/6=1
2/4=2
1/8=8
9/4=2
4/2=2
8/9=1
7/5=1
2/3=1


Exam Summary:
8/1=8 - correct answer
3/8=0 - correct answer
7/6=1 - correct answer
2/4=2 - incorrect answer - the answer is 0
1/8=8 - incorrect answer - the answer is 0
9/4=2 - correct answer
4/2=2 - correct answer
8/9=1 - incorrect answer - the answer is 0
7/5=1 - correct answer
2/3=1 - incorrect answer - the answer is 0
You scored 6/10. Please ask your teacher for help!


Please choose the calculation you want to practice
1) Multiplication
2) Division (quotient)
3) Exit
Please make a selection:
```

(The user's input is 2 to perform divisions. 10 division questions are displayed in the form of x/y= for the user to input answers. The exam summary is displayed marking the user's answers, and displaying the correct answer if the user has answered incorrectly. Message in the case of less than or equal to 7 correct answers is displayed. Automatically directed back to the menu.)

Test Case 6: Selecting 3) Exit

```
Please choose the calculation you want to practice
1) Multiplication
2) Division (quotient)
3) Exit
Please make a selection:
3
Thank you for using the program, goodbye!
Program ended with exit code: 1
```

(The user's input is 3 for exit. Message correctly displayed and the program terminates with 1.)